SACHIN GANPULE

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EXPERIENCE:

Junior Consultant – AI & Automation Engineering | WonderBotz **Aug 2026 - Present**

• Building a **9-stage ETL agent** converting client Process Description Documents into executable ReportBotz rule packages, using deterministic filtering to restrict model inference and a **self-improving skill extraction loop** for increased efficiency.

AI Intern – Data, Cloud & Analytics | Volvo Cars **Jun - Jul 2026**

• Improved customer satisfaction by slashing CX AI agent response latency by **30-60+ seconds**, directly reducing abandonment rates and optimizing compute costs through **semantic caching**.

Designed and built a PoC for a **hybrid RAG pipeline** (Azure AI Search + reranker) to replace the retailer portal search, achieving **95% recall@1** and preserving **Salesforce-driven permission** scoping across webpages, PDFs, Excel, and images.

• Engineered sales forecasting pipeline using **PCA & K-Means clustering to segment volatile monthly demand**, followed by **SARIMAX feature selection and XGBoost modeling** to predict sales target attainment by day 10 of the month, giving leadership an early signal beyond end-of-month projections.

• Translated technical tradeoffs and delivered clear recommendations to both engineering and non-technical stakeholders.

AI/Backend Engineering Intern | Hawl Technologies **Jan - May 2026**

• Constructed a scalable **pgvector**/**HNSW** vector search pipeline, replacing TF-IDF and achieving **92% recall@5**.

• Built **Google Drive MCP integration** with OAuth to support content extraction from docs, sheets, slides, PDFs, and text files.

• Improved backend memory infra by developing a batched embedding API, **reducing 500-memory backfill time by 8x**.

• Extended AI Chrome extension to ChatGLM via **end-to-end scraper and event handshake** for unified conversation capture.

• Standardized development workflows via **Docker** for reproducible testing and **75%** faster engineer onboarding.

RESEARCH & PROJECTS:

Latent Delegation in Coding Agent Harnesses **May 2026 - Present**

• Spearheading novel research on **latent-space inter-agent delegation** within recursive coding frameworks (RecursiveMAS × LangChain Deep Agents), advancing agent observability to target publication at a **ICLR 2027** workshop.

• Designing and executing controlled ablation studies to isolate the efficiency of multi-agent communication, benchmarking critical performance metrics including **success rate, latency, and infrastructure cost**.

Autonomous Trading Floor **Mar 2026**

• Architected a multi-agent system with **4 autonomous agents, 6 MCP servers, and 44 tools** for live market trades.

• Deployed the system as an enterprise-grade SaaS financial planner using a scalable AWS architecture (**Aurora Serverless, Lambda, SQS**) with **LangFuse** for observability.

GrAIdient **Jan - Jun 2025**

• Developed an AI-powered grading tool using OCR to **automate quiz evaluation** from PDFs, reducing grading time by **40%**.

• Architected an end-to-end OCR pipeline, transitioning from Tesseract 5 to a fine-tuned **TrOCR** to boost handwriting recognition accuracy from 40% to **97%**. Integrated a **YOLOv8** model for text detection and automated preprocessing locally.

• Presented GrAIdient at the 2025 Rutgers ECE Capstone Expo; ranked in the **top 15 teams** out of the entire senior class.

EDUCATION:

University of Michigan, Ann Arbor (online) ***Expected Graduation: May 2028***

Master of Applied Data Science (Concentration in AI)

Rutgers University, New Brunswick ***Sep 2021 - May 2025***

Bachelor of Science in Electrical & Computer Engineering

SKILLS:

Programming: Python (NumPy, Pandas, SciPy, Matplotlib) • Javascript/Typescript • SQL • Bash/Zsh • C++ • Java

AI/ML & MLOps: PyTorch • HuggingFace • RAG • SARIMAX • XGBoost • QLoRA/LoRA • RLHF/DPO • OpenAI Agents SDK • LangChain/LangGraph/LangFuse • CrewAI • Docker • Kubernetes/GKE • gVisor • CI/CD • GitHub Actions

Cloud & Dev Tools: AWS(Lambda, SQS, EC2, S3) • React/Next.js • Azure(Foundry, Blob, Entra) • Git • Claude Code

CERTIFICATIONS:

Google Cloud Engineer AI Toolkit: *May 2026* **|** AWS Cloud Practitioner: *Sep 2025* **|** Snowflake GenAI Bootcamp: *June 2025*